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# CUMULATIVE LOGIT ORDINAL REGRESSION FOR SELF-REPORTED HEALTH IN NHANES

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## Abstract

In this report, we analyse self-reported general health using cumulative logit ordinal regression. Self-reported health is usually recorded in ordered categories such as Excellent, Very good, Good, Fair and Poor. Because these categories have a clear order, we used an ordinal regression approach rather than treating the response as a numerical variable.

We used data from the National Health and Nutrition Examination Survey (NHANES), focusing on survey cycles 2009–2012. Our response variable was self-reported general health, and our predictors were age, body mass index (BMI), gender and current smoking status. After selecting the required variables and keeping complete cases, our final modelling dataset contained 2,885 observations.

We fitted both parallel and non-parallel cumulative logit models. At the start, we expected the non-parallel model to be more useful because it is more flexible and allows predictor effects to vary across thresholds. However, when we fitted it, the output became much more complicated and less stable to interpret. The non-parallel model gave only a very small reduction in residual deviance compared with the parallel model, and its log-likelihood was not available in the fitted output. This was one of the main points we learned from the analysis: a more flexible model is not automatically a better final model.

Overall, we found that cumulative logit ordinal regression was suitable for this ordered health outcome. The parallel model was selected as the main model because it was simpler, more stable and easier to explain. The non-parallel model was still useful as a sensitivity check, because it showed that relaxing the parallel assumption did not meaningfully improve the model for our data.

**Project team:** Manasvi Saite and Kaushal Mayur  
**Supervisor:** Dr. Paul Wilson

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## 1 Acknowledgements

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## 2 Introduction

Self-reported health is a common measure used in public health and health survey research. It is usually collected by asking people to describe their general health using categories such as Excellent, Very good, Good, Fair and Poor. Although the measure is subjective, it is useful because it reflects how people judge their own health condition. This judgement may be influenced by physical health, lifestyle, long-term illness, mental wellbeing and social factors.

Our main aim is to analyse self-reported general health using data from the National Health and Nutrition Examination Survey (NHANES). Our analysis focuses on selected demographic and health-related predictors: age, body mass index (BMI), gender and current smoking status. The response variable is self-reported health, and the aim is to understand how the selected predictors are associated with the ordered health categories.

The research question for this project is:

**How are age, BMI, gender and current smoking status associated with ordered self-reported general health in the NHANES data?**

Because we worked on this as a two-student project, we wanted the report to show our thinking, not just the final model output. For that reason, we included both the overall model results and a worked example for one person.

### 2.1 Project aims

The aims of this project are:

1. to describe the ordered self-reported health outcome in the NHANES modelling dataset;
2. to examine the relationship between self-reported health and age, BMI, gender and current smoking status;
3. to fit and compare parallel and non-parallel cumulative logit ordinal regression models;
4. to interpret estimates, odds ratios, p-values and predicted probabilities in clear applied terms;
5. to use an individual prediction example to explain how fitted ordinal regression models assign probabilities to health categories;
6. to provide project code and the interactive predicted probability visualisation through the GitHub repository.

Together, these aims helped us keep the project focused on both the statistical model and the practical meaning of the results.

The response variable is ordered, so ordinal regression is suitable for this research question. Excellent is the highest health category and Poor is the lowest health category. For example, one cannot assume that the difference between Excellent and Very good is the same as the difference between Fair and Poor. Therefore, the outcome should not be treated as a continuous numerical variable. This follows the general idea of ordered regression models, where the response has ranked categories and the modelling approach should preserve that order (McCullagh 1980; Agresti 2010; Fullerton and Xu 2016; Course Material 2026a).

We used cumulative logit ordinal regression. Two versions of the model are fitted: the parallel model and the non-parallel model. The parallel model assumes that the predictor effects are the same across cumulative thresholds. The non-parallel model allows the predictor effects to differ across thresholds. The two models are compared using model output, residual deviance, fit statistics where available, predicted probabilities and interpretability.

We organised the report as follows. First, ordinal data and self-reported health are described. Next, the NHANES dataset and selected variables are introduced. Exploratory data analysis is then used to study the

outcome and predictors. The methodology section explains cumulative logit ordinal regression, the parallel model and the non-parallel model. The results section presents fitted model outputs, odds ratios, p-values, predicted probabilities and an example prediction for a particular person. Finally, the report discusses the model choice, limitations and possible future work. The report was written in RMarkdown using the `rticles` arXiv article format, with references managed through a separate BibTeX file (Allaire, Xie, Dervieux, et al. 2024; Allaire, Xie, Foundation, et al. 2024; Xie et al. 2020).

### 3 Project Team and Working Approach

We completed this project by two students, Manasvi Saite and Kaushal Mayur, under the supervision of Dr. Paul Wilson. We worked together on the report structure, model interpretation and final presentation of results. We used the GitHub repository to keep the project files together, including the RMarkdown report, code file, bibliography and interactive HTML outputs.

Our main aim was to understand the output, not just run the model. At first, the cumulative logit model looked complicated because the output gives intercepts, estimates, odds ratios and cumulative probabilities. We therefore decided to include a step-by-step individual prediction example in the report. This helped us explain how the model moves from estimates to logits, then to cumulative probabilities and finally to category probabilities.

The non-parallel model was included to check whether relaxing the proportional odds assumption would add useful information beyond the parallel model. Although this specification allows predictor effects to vary across thresholds, the fitted results did not show a meaningful practical gain. The reduction in residual deviance was very small, the output became more detailed, and the log-likelihood was not available in the fitted non-parallel output. For these reasons, the non-parallel model was retained as a sensitivity check rather than used as the main final model.

The final model choice was based on several pieces of evidence rather than on model complexity alone. The comparison considered residual deviance, available fit statistics, predicted probability patterns and the consistency of the substantive conclusions. Since the more flexible model did not materially change the findings, the parallel model was selected as the main model for reporting.

### 4 Background and Related Work

Self-reported health is widely used in health research because it gives a summary of how individuals assess their own health status. It is not a clinical diagnosis, but it is still useful because it captures a person's overall judgement of health. This judgement may reflect physical health, symptoms, lifestyle, age, body weight, smoking behaviour and other personal factors.

The outcome is naturally ordinal because the response categories have a clear order. Ordered categorical outcomes require methods that preserve this ranking. Treating the outcome as continuous would ignore the categorical nature of the response, while treating it as nominal would ignore the ordering. Cumulative logit ordinal regression provides a suitable middle ground because it uses the order of the categories without assuming that they are measured on a continuous scale (McCullagh 1980; Agresti 2010).

Previous methodological work on ordered regression models distinguishes between parallel, partial and non-parallel forms (Fullerton and Xu 2016; Williams 2006). The parallel model provides one main effect for each predictor, while the non-parallel model allows predictor effects to change across thresholds. Both approaches were used so that the simpler model could be checked against a more flexible alternative.

### 5 Ordinal Data

Ordinal data are categorical data with a meaningful order. In our project, the outcome variable is self-reported general health. The categories are:

**Excellent, Very good, Good, Fair and Poor.**

These categories clearly have a hierarchy. Excellent represents the best health category and Poor represents the worst health category. The category order is important, but the response is still categorical. This is the main reason why ordinary linear regression is not suitable for the outcome variable.

A simple way to understand ordinal data is to think of the categories as labels with order. Numbers such as 1 to 5 can be used to label the categories, but these numbers are not the same as regular measurements. If Excellent is coded as 1 and Poor is coded as 5, it does not mean that Poor is five times worse than Excellent. The numbers show ranking, not exact measurement.

Ordinal regression is more suitable because it keeps the order of the categories without treating the response as continuous. This allows the analysis to use the information that Excellent is better than Very good, Very good is better than Good, and so on. Cumulative logit ordinal regression is a widely used method for ordered categorical outcomes (McCullagh 1980; Agresti 2010).

In our project, ordinal regression is used to model the relationship between self-reported health and four predictors: age, BMI, gender and current smoking status. The model helped us explain whether these predictors are associated with better or poorer self-reported health.

## 6 Data Source and Variables

NHANES is the National Health and Nutrition Examination Survey. It is a health survey of people in the United States conducted by the National Center for Health Statistics, which is part of the Centers for Disease Control and Prevention. NHANES collects information using questionnaire responses, physical examination measurements and health-related variables (Centers for Disease Control and Prevention, National Center for Health Statistics 2024a).

We used the NHANES data available through the R package NHANES (Pruim 2016). The data used in this project are from survey cycles 2009–2012. These data include the variables needed for the analysis: self-reported general health, age, BMI, gender and current smoking status.

The dependent variable is self-reported general health. In the dataset, this is recorded as ordered categories. The categories used in this analysis are Excellent, Very good, Good, Fair and Poor. The predictors used in the model are:

- **Age:** numerical predictor.
- **BMI:** numerical predictor representing body mass index.
- **Gender:** categorical predictor.
- **Smoking:** categorical Yes/No variable showing current smoking status.

The data preparation was completed in R using the `dplyr` workflow from the tidyverse (R Core Team 2024; Wickham et al. 2019). The analysis workflow created the ordered health outcome, converted the predictors into the correct formats, selected the variables used in the model, removed incomplete cases and dropped unused factor levels. The full analysis code is documented in the GitHub repository.

After selecting the required variables and keeping complete cases, the modelling dataset contained participants aged 20 to 80 years. This age range was observed in the selected modelling dataset; it was not added as a separate manual restriction in the analysis. The model therefore examines how age is associated with self-reported health across the available age range in the data.

Table 1 describes how the variables were used in the analysis.

Table 1: Variables used in the ordinal regression analysis.

| Variable | Role      | Type                | Coding_or_description                  |
|----------|-----------|---------------------|----------------------------------------|
| Health   | Outcome   | Ordered categorical | Excellent, Very good, Good, Fair, Poor |
| Age      | Predictor | Numeric             | Age in years                           |
| BMI      | Predictor | Numeric             | Body mass index                        |
| Gender   | Predictor | Categorical         | Female or male                         |
| Smoking  | Predictor | Categorical         | Current smoking status: No or Yes      |

Table 1 shows that the outcome variable is ordered categorical, while the predictors include both numerical and categorical variables. This mix of variables is appropriate for cumulative logit ordinal regression.

Missing values were checked for the selected variables before fitting the models. This step was included as a data quality check, even though no major missing-data cleaning was required in the final modelling dataset. Table 2 presents the missing-value summary for the selected variables.

Table 2: Missing-value summary for selected modelling variables.

| Variable | Missing_Count | Missing_Percent |
|----------|---------------|-----------------|
| Health   | 0             | 0               |
| Age      | 0             | 0               |
| Gender   | 0             | 0               |
| BMI      | 0             | 0               |
| Smoking  | 0             | 0               |

As shown in Table 2, the selected variables Health, Age, Gender, BMI and Smoking had zero missing values in the final modelling dataset. This showed that the models were fitted using complete cases for all variables included in the analysis.

Table 3 summarises the final modelling dataset. It shows the number of complete cases, the observed age range, the mean age, the mean BMI and the number of outcome categories.

Table 3: Summary of the final modelling dataset.

| Item               | Value   |
|--------------------|---------|
| Complete cases     | 2885    |
| Age range          | 20 - 80 |
| Mean age           | 49.25   |
| Mean BMI           | 28.56   |
| Outcome categories | 5       |

Table 3 shows that the final dataset contains 2,885 complete cases. The average age is approximately 49.25 years and the average BMI is approximately 28.56 kg/m<sup>2</sup>. These values provide a basic description of the sample before fitting the ordinal regression models.

## 7 Exploratory Data Analysis

Before fitting the ordinal regression models, exploratory data analysis was carried out to understand the structure of the data. This step helped examine the distribution of the outcome variable, the distribution of the selected predictors, and the visible patterns between predictors and self-reported health. We completed the analysis in R (R Core Team 2024). The graphs were created using the `ggplot2` package (Wickham 2016), and tables were produced using `knitr` (Xie 2015).

### 7.1 Frequency tables for categorical variables

Table 4 shows the frequency of each self-reported health category. This table is important because ordinal regression models can be affected by very small categories.

Table 4: Frequency of self-reported health categories.

| Health category | Count |
|-----------------|-------|
| Excellent       | 213   |
| Very good       | 849   |
| Good            | 1228  |
| Fair            | 496   |
| Poor            | 99    |

As shown in Table 4, most participants reported their health as Good or Very good, while fewer participants reported Poor health. We considered this pattern important because the Poor category has fewer observations than the middle categories.

Table 5 shows the gender distribution in the modelling dataset.

Table 5: Frequency of gender categories.

| Gender | Count |
|--------|-------|
| female | 1236  |
| male   | 1649  |

Table 5 confirms that both female and male participants are represented in the modelling dataset. Gender is therefore included as a categorical predictor in the model.

Table 6 shows the distribution of current smoking status.

Table 6: Frequency of current smoking status.

| Current smoking | Count |
|-----------------|-------|
| No              | 1581  |
| Yes             | 1304  |

As shown in Table 6, there are more non-smokers than current smokers. This is useful to know because the smoking coefficient in the model compares current smokers with non-smokers while holding other predictors constant.

## 7.2 Graphical exploration

Figure 1 shows the distribution of self-reported health. This figure provides a visual version of Table 4.

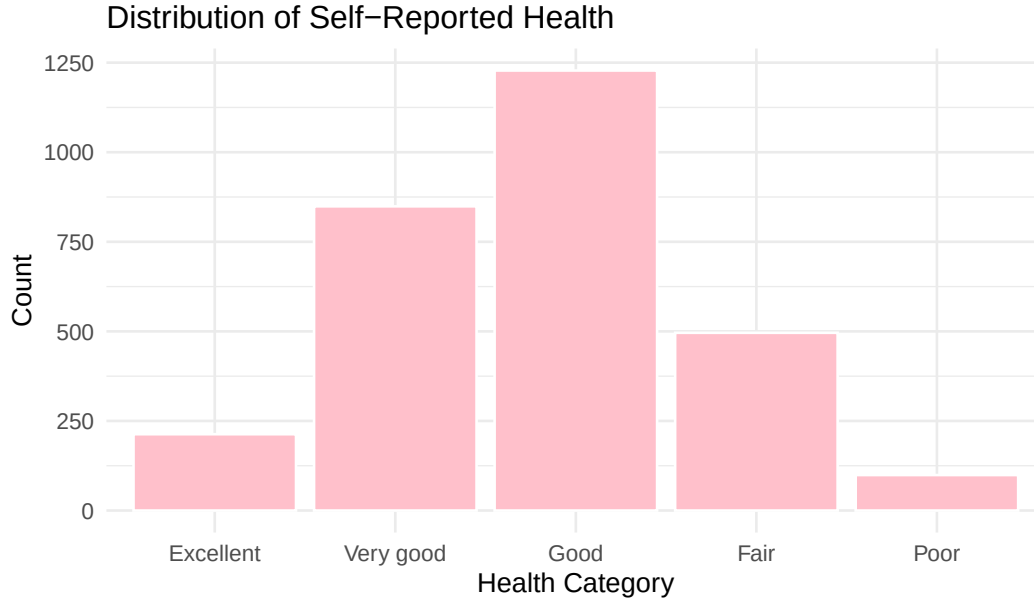


Figure 1: Distribution of self-reported health categories.

Figure 1 shows that most participants are in the Good and Very good categories. The smaller number of Poor cases is important because very small categories can make flexible models harder to estimate reliably.



Figure 2 shows the gender distribution in the modelling dataset.

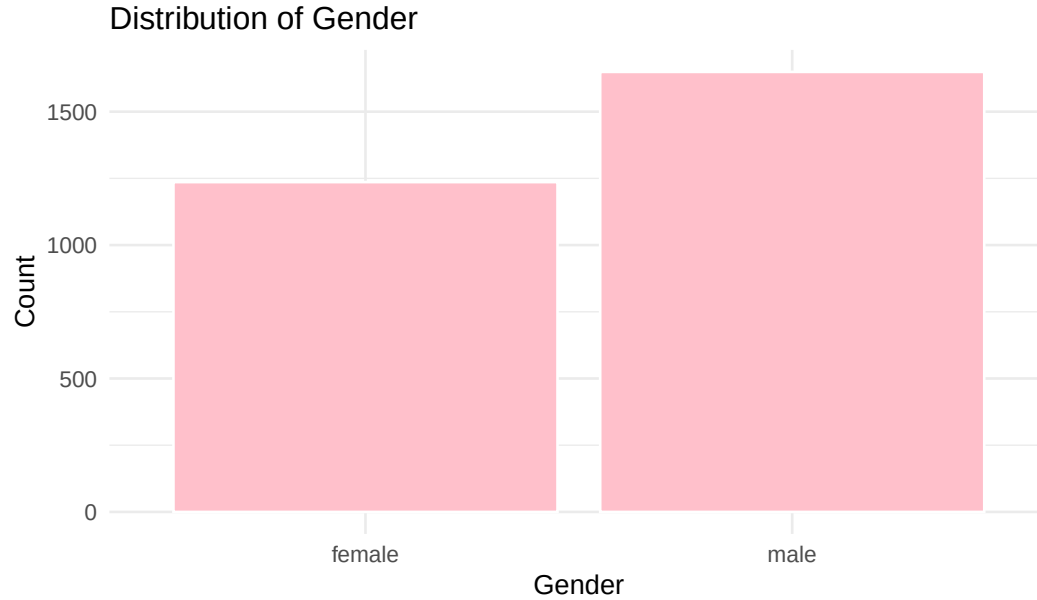


Figure 2: Distribution of gender in the modelling dataset.

Figure 2 shows that both gender categories are present in the data. This supported our use of gender as a categorical predictor in the ordinal regression model.

Figure 3 shows the distribution of current smoking status.

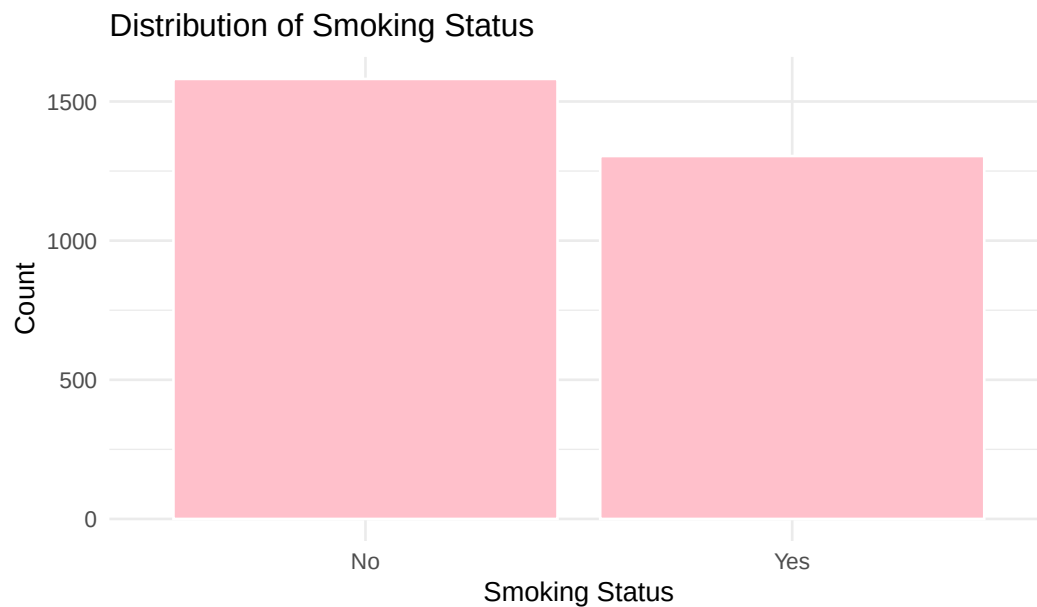


Figure 3: Distribution of smoking status.

Figure 3 shows that non-smokers are more common than current smokers. The model uses this variable to estimate whether current smoking is associated with poorer self-reported health after adjusting for age, BMI and gender.

Figure 4 shows the age distribution available in the selected modelling dataset.

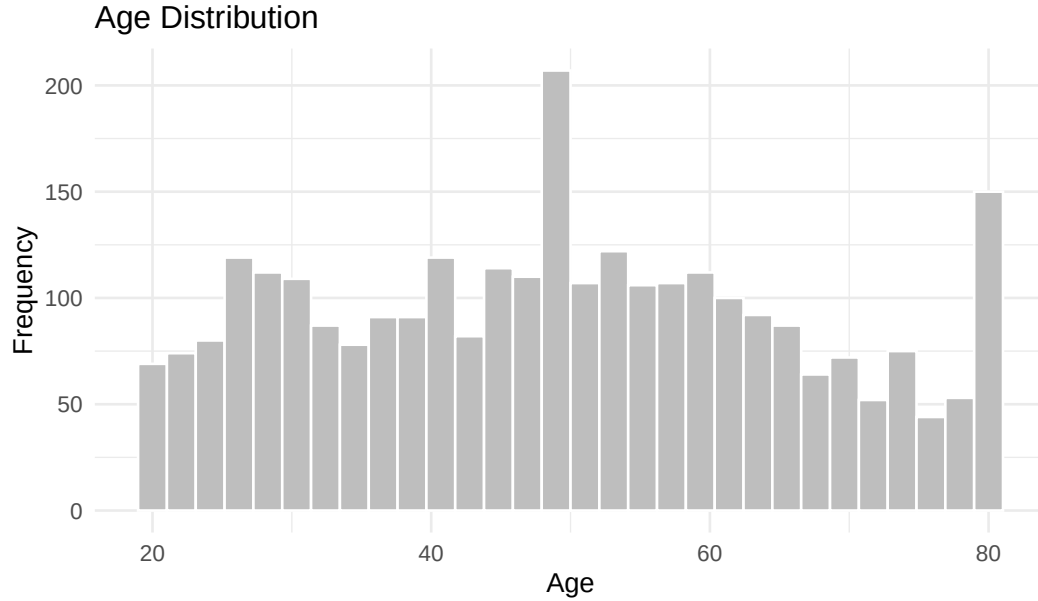


Figure 4: Age distribution of participants in the modelling dataset.

Figure 4 shows the observed age range in the modelling dataset. Age was included as a numerical predictor because one of the aims of the project was to understand how self-reported health is associated with age. Age was also used later in the predicted probability plots.

Figure 5 shows the BMI distribution.

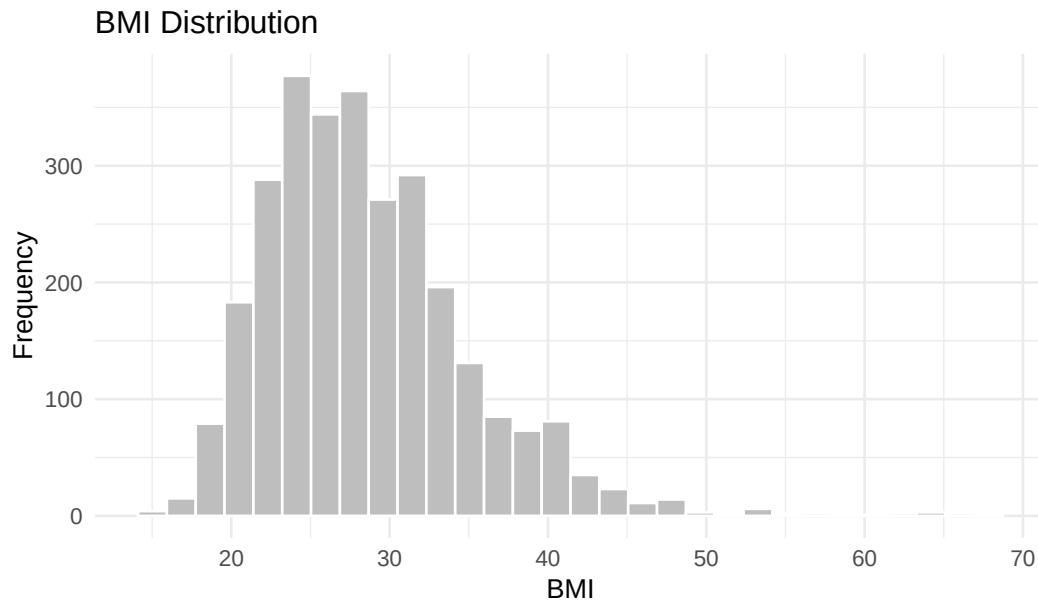


Figure 5: BMI distribution of participants in the modelling dataset.

Figure 5 shows that most BMI values are around the mid-to-high 20s, with some higher values. BMI was included as a continuous predictor because it is a common health-related measure and may be associated with self-reported health.

Figure 6 compares age across the five self-reported health categories.

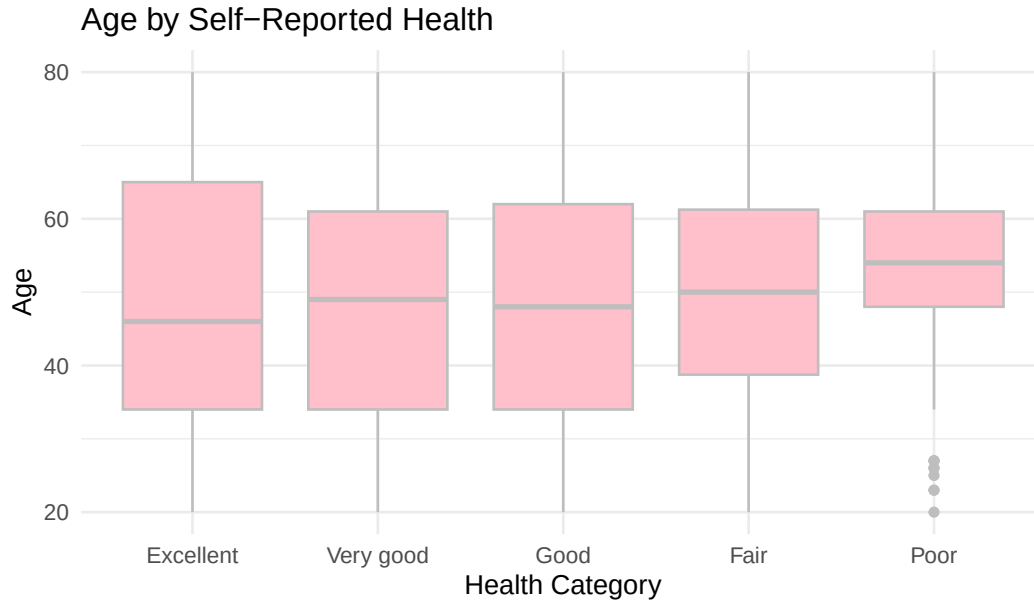


Figure 6: Age by self-reported health category.

Figure 6 suggests that older participants are more common in poorer health categories. This gives a visual reason for including age in the model.

Figure 7 compares BMI across the five self-reported health categories.

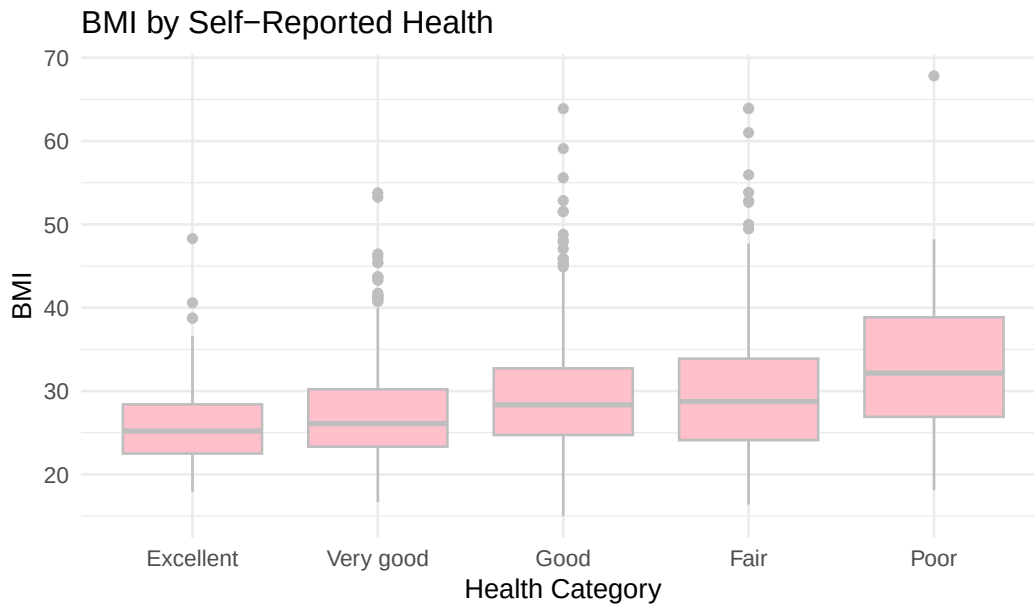


Figure 7: BMI by self-reported health category.

Figure 7 shows that BMI tends to be higher in the poorer self-reported health categories. This supports the inclusion of BMI as a predictor in the ordinal regression model.

Overall, the exploratory analysis shows that the outcome is ordered and that the selected predictors have visible relationships with self-reported health. This supported our decision to use cumulative logit ordinal regression.

## 8 Methods

We followed a standard applied modelling workflow. First, the variables required for the project were selected from the NHANES data. The response variable was recoded as an ordered factor with five health categories. The predictors were prepared as numerical or categorical variables depending on their role in the model. Complete cases were then used for the final modelling dataset.

Second, exploratory data analysis was used to understand the distribution of the outcome and predictors. This included frequency tables for categorical variables and graphical summaries for health, gender, smoking status, age and BMI. Boxplots were used to compare age and BMI across the ordered health categories.

Third, cumulative logit ordinal regression models were fitted. The parallel model was fitted as the main model because it gives one interpretable effect for each predictor. We fitted the non-parallel model as a sensitivity check because it allows threshold-specific effects. Model comparison was based on residual deviance, available fit statistics, predicted probability patterns and interpretability.

Finally, predicted probabilities were calculated for both general age-based profiles and an illustrative individual profile. This helped translate the fitted model from log-odds and odds ratios into probabilities that are clearer to report.

## 9 Cumulative Logit Ordinal Regression

Ordered regression models are used when the response variable has categories with a natural ranking. The literature describes several approaches for ordered outcomes, including cumulative models, stage models and adjacent category models (Fullerton and Xu 2016; Course Material 2026a). In this project, the cumulative logit model was chosen because the aim was to model the ordered self-reported health outcome while keeping the interpretation clear and suitable for the project question.

Cumulative logit ordinal regression is used when the outcome variable has ordered categories. In our project, the outcome is self-reported health, ordered from Excellent to Poor. Since the response is categorical and ordered, cumulative logit ordinal regression is more suitable than ordinary linear regression.

The main idea of the model is to use Odds ratio. Instead of modelling only one health category at a time, the model considers the probability of being at or below a particular health threshold. For example, it can compare the odds of being in Excellent health versus all other categories, then Excellent or Very good versus the remaining categories, and so on.

For five ordered health categories, there are four cumulative thresholds. These thresholds split the ordered response into cumulative comparisons. In this project, the comparisons can be understood as moving across the health scale from the best category towards the poorest category. This is helpful because the model uses the full ordering of the response instead of ignoring the order or treating the categories as ordinary numbers.

For an ordered response variable  $Y$ , the cumulative logit model can be written as:

$$\log \left( \frac{P(Y \leq j)}{P(Y > j)} \right) = \alpha_j + \beta X$$

where  $Y$  is the ordered self-reported health outcome,  $j$  represents a health category threshold,  $\alpha_j$  is the threshold-specific intercept,  $X$  represents the predictors and  $\beta$  represents the predictor effect.

In our project, the predictors are age, BMI, gender and current smoking status. The model estimated how these predictors are associated with the cumulative odds of reporting better or poorer self-reported health. The cumulative logit model is commonly used for ordinal response variables and can be fitted under a proportional odds, or parallel slopes, assumption (McCullagh 1980; Agresti 2010).

The coefficients from the model are estimated on the log-odds scale. Log-odds are useful for modelling, but they are not easy to explain directly. Therefore, the coefficients are converted into odds ratios by exponentiating them:

$$\text{Odds Ratio} = \exp(\beta)$$

An odds ratio helps explain the direction and strength of the relationship between a predictor and the outcome. If the odds ratio is greater than 1, the odds increase. If the odds ratio is less than 1, the odds decrease. In this project, because of the ordering and coding direction of the outcome, odds ratios below 1 indicate lower cumulative odds of being in better self-reported health categories.

For example, if the odds ratio for current smoking is below 1, it means that current smokers have lower odds of reporting better health compared with non-smokers, after adjusting for age, BMI and gender. This does not mean the probability directly decreases by the same percentage. It means the odds are lower on the odds scale.

We fitted the cumulative logit models using the **VGAM** package and the `vglm()` function in R (Yee 2010, 2015, 2024). Two model forms were fitted: the parallel model and the non-parallel model. These two models are discussed in the next section.

## 10 Parallel and Non-Parallel Models

The cumulative logit ordinal regression model can be fitted in two main forms: the parallel model and the non-parallel model. Both models are designed for ordered categorical outcome variables, but they differ in the assumptions they make about the predictor effects.

In this project, the ordered outcome variable is self-reported health. The categories are Excellent, Very good, Good, Fair and Poor. The predictors are age, BMI, gender and current smoking status. Since the outcome has a natural order, cumulative logit ordinal regression is appropriate. However, it is still necessary to decide whether the predictor effects should be assumed to be the same across all cumulative thresholds or whether they should be allowed to vary.

The parallel model assumes that each predictor has the same effect across all cumulative thresholds. This assumption is also known as the proportional odds assumption. Under this assumption, the model estimates one coefficient for each predictor. This means that age has one estimated effect, BMI has one estimated effect, gender has one estimated effect and smoking has one estimated effect across the ordered health outcome.

For example, the effect of BMI is assumed to be consistent across the cumulative thresholds. This made the model easier for us to interpret because each predictor has only one overall odds ratio. The parallel model is therefore useful when the goal is to obtain a clear summary of the relationship between predictors and the ordered health outcome.

We fitted the parallel cumulative logit model in R using the `vglm()` function from the **VGAM** package (Yee 2010, 2015, 2024). The model specification is documented in the project code repository, while the fitted results are reported in the tables below.

The argument `parallel = TRUE` tells R to fit the proportional odds version of the cumulative logit model. This means the same slope coefficient is used across the cumulative logits.

The non-parallel model relaxes the proportional odds assumption. Instead of forcing the effect of each predictor to be the same across all thresholds, the non-parallel model allows the effect of a predictor to differ from one threshold to another. This made the model more flexible.

We fitted the non-parallel model using the same variables as the parallel model, with the model option `parallel = FALSE` in the project code.

The argument `parallel = FALSE` allows separate coefficients to be estimated for each cumulative threshold. This increases model flexibility, but it also increases model complexity. The output becomes longer because each predictor can have multiple threshold-specific coefficients.

### 10.1 What we observed when fitting the non-parallel model

When we fitted the non-parallel model, we expected it to improve the analysis because it allowed age, BMI, gender and smoking to have different effects at each cumulative threshold. In theory, this is useful because the effect of a predictor may not be the same when comparing the best health category with the rest and when comparing the poorest health category with the rest.

However, in practice, the model became much more complex to report. Instead of one coefficient for age, one for BMI, one for gender and one for smoking, the model gave threshold-specific coefficients. This made the output long and less direct. We also noticed that the fitted output did not provide a usable log-likelihood value, while the parallel model produced a stable log-likelihood, AIC and BIC. This made the parallel model easier to report clearly.

The non-parallel model did reduce the residual deviance slightly. However, the reduction was only about 1.068 compared with a residual deviance above 7400. We found this improvement too small to justify using the more complicated model as the main result. For this reason, we kept the non-parallel model in the report, but used it mainly as a sensitivity check.

Both models were included because they provide different information. The parallel model gives a simpler and more interpretable summary of predictor effects. The non-parallel model is useful as a sensitivity check because it examines whether allowing threshold-specific effects gives a noticeably better model.

A more flexible model is not always the best model. A model may have a slightly better fit but may also be more complex to report, less stable or less useful for interpretation. Therefore, the final model choice in this project considered both statistical fit and practical interpretation.

## 11 Model Estimation and Fit Measures

The cumulative logit models in this project are fitted using maximum likelihood estimation. Maximum likelihood estimation chooses the parameter values that make the observed data most likely under the model. In our interpretation, the model searches for the estimates that best match the observed pattern of self-reported health categories.

Likelihood-based estimation is useful for ordinal regression because the outcome is categorical. Unlike ordinary linear regression, where the residual sum of squares is commonly used, ordinal regression uses likelihood-based measures such as log-likelihood, residual deviance, AIC and BIC. These measures help describe how well the model fits the observed data.

The log-likelihood gives a measure of how well the fitted model explains the observed response pattern. A larger log-likelihood is usually preferred when comparing models fitted to the same data. However, log-likelihood alone does not penalise model complexity. A more complex model can often fit the data slightly better simply because it has more parameters.

AIC and BIC are therefore useful because they consider both model fit and model complexity (Course Material 2026b; Burnham and Anderson 2002). AIC adds a penalty for the number of parameters in the model. BIC also penalises complexity, and usually applies a stronger penalty when the sample size is large. Lower AIC and BIC values generally indicate a more favourable model, but the final decision should also consider interpretability and stability.

Residual deviance is another important fit measure. A smaller residual deviance usually indicates a better-fitting model. However, the size of the reduction must be interpreted carefully. In this project, the non-parallel model had a slightly lower residual deviance than the parallel model, but the improvement was very small. Because the non-parallel model was much more complex, the very small reduction did not justify using it as the main model.

We considered this important for the final model decision. For us, the aim was not just to fit the most flexible model, but to choose a model that answered the research question clearly. The parallel model gives one main effect for each predictor. This makes the interpretation easier and more useful for explaining how age, BMI, gender and smoking are associated with self-reported health.

## 12 Fitted Model Results

This section presents the fitted model outputs. The focus is on coefficient estimates, odds ratios, p-values, residual deviance and predicted probabilities. Our aim is not only to report the model output, but also to explain what the values mean in the context of self-reported health.

### 12.1 Parallel model output

The parallel model was fitted first. The fitted model contains threshold estimates and predictor estimates. The main predictor results are extracted and presented in Table 7.

The main predictor coefficients from the parallel model were extracted and converted into odds ratios. P-values were also included to identify which predictors showed statistical evidence of association with self-reported health.

Table 7: Parallel model coefficients, odds ratios, confidence intervals and p-values.

| Predictor      | Estimate | SE     | z       | OR    | 95% CI         | p      |
|----------------|----------|--------|---------|-------|----------------|--------|
| Age            | -0.0118  | 0.0022 | -5.379  | 0.988 | 0.984 to 0.993 | <0.001 |
| BMI            | -0.0733  | 0.0056 | -13.038 | 0.929 | 0.919 to 0.94  | <0.001 |
| Male           | -0.1533  | 0.0696 | -2.203  | 0.858 | 0.748 to 0.983 | 0.028  |
| Current smoker | -0.8398  | 0.0752 | -11.168 | 0.432 | 0.373 to 0.5   | <0.001 |

Table 7 shows the main predictor results from the parallel cumulative logit model. The Estimate column gives the coefficient on the log-odds scale. The Odds Ratio column gives the exponentiated coefficient, which is clearer to report. The p-value column shows whether there is statistical evidence that the predictor is associated with self-reported health after adjusting for the other predictors.

The age estimate is negative, meaning that increasing age is associated with lower cumulative odds of better self-reported health. The odds ratio for age is slightly below 1. This means that each one-year increase in age is associated with a small decrease in the odds of better self-reported health, holding BMI, gender and smoking status constant.

The BMI estimate is also negative. The odds ratio for BMI is below 1, meaning that higher BMI is associated with lower odds of better self-reported health, after adjusting for the other variables. This matches the exploratory analysis, where BMI tended to be higher among poorer health categories.

The male coefficient compares males with the reference gender category. The odds ratio below 1 suggests that males have lower cumulative odds of better self-reported health compared with the reference category, after adjusting for age, BMI and smoking.

The smoking coefficient is the strongest effect in the parallel model. The odds ratio for current smoking is clearly below 1. This means current smokers have lower odds of better self-reported health compared with non-smokers, after adjusting for age, BMI and gender. This is one of the most important findings of the model.

The hypothesis testing interpretation follows the usual rule: if the p-value is below 0.05, there is statistical evidence that the predictor is associated with self-reported health. In this model, age, BMI and smoking show statistical evidence of association, while gender is weaker compared with the other predictors. This matches the manual interpretation notes used during the project work (Project Calculation Notes 2026).

The fit statistics for the parallel model are shown in Table 8.

Table 8: Parallel model fit statistics.

| Measure           | Value     |
|-------------------|-----------|
| Residual deviance | 7418.732  |
| Log-likelihood    | -3709.366 |
| AIC               | 7434.732  |

| Measure | Value    |
|---------|----------|
| BIC     | 7482.470 |

Table 8 shows the residual deviance, log-likelihood, AIC and BIC for the parallel model. These values are used to describe overall model fit. The AIC and BIC are useful because they consider both fit and model complexity. Lower values are generally preferred when comparing models fitted to the same data, although interpretation and stability must also be considered.

## 12.2 Non-parallel model output

The non-parallel model was then fitted. This model gives separate threshold-specific coefficients, and the extracted results are presented in Table 9.

The non-parallel coefficient table is longer because each predictor has coefficients for different cumulative thresholds.

Table 9: Non-parallel model coefficient output.

| Term          | Estimate | Std_Error | z_value  | Odds_Ratio | p_value_formatted |
|---------------|----------|-----------|----------|------------|-------------------|
| (Intercept):1 | 0.3888   | 0.4120    | 0.9436   | 1.4752     | 0.345             |
| (Intercept):2 | 2.3647   | 0.2463    | 9.6021   | 10.6411    | <0.001            |
| (Intercept):3 | 4.2791   | 0.2933    | 14.5916  | 72.1742    | <0.001            |
| (Intercept):4 | 7.4781   | 0.4824    | 15.5031  | 1768.9005  | <0.001            |
| Age:1         | -0.0068  | 0.0042    | -1.6202  | 0.9932     | 0.105             |
| Age:2         | -0.0097  | 0.0025    | -3.8819  | 0.9904     | <0.001            |
| Age:3         | -0.0147  | 0.0030    | -4.9258  | 0.9854     | <0.001            |
| Age:4         | -0.0223  | 0.0056    | -3.9593  | 0.9779     | <0.001            |
| BMI:1         | -0.0755  | 0.0131    | -5.7393  | 0.9273     | <0.001            |
| BMI:2         | -0.0697  | 0.0069    | -10.0572 | 0.9326     | <0.001            |
| BMI:3         | -0.0607  | 0.0069    | -8.8427  | 0.9411     | <0.001            |
| BMI:4         | -0.1025  | 0.0081    | -12.7178 | 0.9026     | <0.001            |
| Gendermale:1  | -0.1101  | 0.1372    | -0.8025  | 0.8958     | 0.422             |
| Gendermale:2  | -0.1284  | 0.0797    | -1.6113  | 0.8795     | 0.107             |
| Gendermale:3  | -0.1951  | 0.0939    | -2.0780  | 0.8227     | 0.038             |
| Gendermale:4  | 0.2106   | 0.1822    | 1.1559   | 1.2344     | 0.248             |
| SmokingYes:1  | -0.8092  | 0.1524    | -5.3087  | 0.4452     | <0.001            |
| SmokingYes:2  | -0.8026  | 0.0861    | -9.3196  | 0.4482     | <0.001            |
| SmokingYes:3  | -0.7970  | 0.1003    | -7.9434  | 0.4507     | <0.001            |
| SmokingYes:4  | -0.5184  | 0.1961    | -2.6429  | 0.5955     | 0.008             |

Table 9 shows the non-parallel model output. Unlike the parallel model, the non-parallel model provides threshold-specific effects. This means that age, BMI, gender and smoking can have different estimated effects at different cut-points of the outcome.

The advantage of the non-parallel model is flexibility. It can capture situations where a predictor has a stronger effect at one part of the health scale than another. For example, smoking may affect the probability of being in the best health categories differently from the probability of being in the poorest health category.

The disadvantage is that the output is more complex to report. Instead of one odds ratio for each predictor, there are several odds ratios for each predictor. This made the model less simple and less suitable as the main interpretation model, unless the improvement in fit is clearly meaningful.

## 12.3 Model deviance comparison

Residual deviance was used to compare the parallel and non-parallel models. A lower residual deviance indicates better fit, but the size of the improvement must be considered. In likelihood-based modelling, deviance compares the fitted model with a saturated model, and values closer to zero indicate better fit relative to the best possible model (Course Material 2026b; Agresti 2015).



Table 10: Residual deviance comparison between the parallel and non-parallel models.

| Model              | Residual_Deviance |
|--------------------|-------------------|
| Parallel model     | 7418.732          |
| Non-parallel model | 7417.663          |
| Difference         | 1.069             |

Table 10 shows that the non-parallel model has a slightly smaller residual deviance than the parallel model. The difference is only 1.068. Since both deviance values are above 7400, this is a very small improvement. This means that the non-parallel model does not provide a meaningful improvement in fit compared with the extra complexity it adds.

### 13 Understanding Estimates, Odds Ratios and P-values

This section explains the main model output in simple terms. It is included because model coefficients, odds ratios and p-values are often misunderstood.

#### 13.1 Estimates

The estimate represents the model coefficient and is reported on the log-odds scale. A positive estimate suggests that the predictor is associated with an increase in the cumulative log-odds, while a negative estimate suggests a decrease in the cumulative log-odds.

In our project, the outcome categories are ordered from Excellent to Poor. Because of this coding direction, negative predictor estimates generally indicate lower cumulative odds of being in better health categories. For example, a negative smoking estimate suggests that current smokers are less likely to report better health compared with non-smokers, after adjusting for the other predictors.

The estimate is important because it gives the direction of the association. However, log-odds are less direct to report directly, so odds ratios are usually reported.

#### 13.2 Odds ratios

An odds ratio is calculated by exponentiating the estimate:

$$\text{Odds Ratio} = \exp(\text{Estimate})$$

An odds ratio of 1 means no difference in odds compared with the reference group or reference value. An odds ratio greater than 1 means the odds are higher than the reference, while an odds ratio less than 1 means the odds are lower than the reference.

For example, if the smoking odds ratio is 0.432, this means current smokers have 0.432 times the cumulative odds of better self-reported health compared with non-smokers, after adjusting for age, BMI and gender. Another way to explain this is that current smokers have approximately  $1 - 0.432 = 0.568$ , or 56.8%, lower odds of better self-reported health than non-smokers, after adjusting for age, BMI and gender. This is an odds interpretation, not a direct probability interpretation.

#### 13.3 P-values

A p-value helps assess whether there is statistical evidence that a predictor is associated with the outcome. A small p-value means the observed association is unlikely to be due to random variation alone, assuming the model is correct. A common threshold is 0.05.

In our project, p-values below 0.05 are treated as evidence that the predictor is associated with self-reported health after adjusting for the other variables. However, a p-value does not measure the size of the effect. For effect size, the odds ratio is more useful.

A p-value also does not prove causation. We used observational survey data, so the results should be interpreted as associations rather than causal effects.

## 14 Practical Interpretation of the Main Predictors

The model results can also be understood in practical terms. This section explains the main predictors in a way that connects the numerical output to the research question.

Age was included because health status often changes across the life course. In the parallel model, the age coefficient was negative and statistically significant. This means that, after adjusting for BMI, gender and smoking status, older age was associated with lower odds of better self-reported health. The effect for one year of age is small, but the effect becomes more meaningful across a larger age difference. For example, a 20-year increase in age would have a stronger practical meaning than a one-year increase.

BMI was also included as a health-related numerical predictor. The BMI coefficient was negative and statistically significant. This suggests that higher BMI was associated with lower odds of better self-reported health, after adjusting for the other predictors. This result matches the exploratory analysis, where BMI tended to be higher in poorer health categories.

Current smoking status was one of the strongest predictors in the model. The odds ratio for current smokers was well below 1. This means that current smokers had lower odds of reporting better health compared with non-smokers, after controlling for age, BMI and gender. The smoking result is also easy to explain in public health terms because smoking is a well-known lifestyle factor connected with poorer health.

Gender was included as a categorical predictor. The gender coefficient was weaker than age, BMI and smoking. Although the p-value showed some evidence of association in the parallel model, the size of the effect was smaller than the smoking effect. Therefore, gender is included in the model, but it is not the main driver of the findings.

The overall pattern is consistent across the analysis. The exploratory plots showed that age and BMI had visible patterns across health categories. The model then confirmed that age, BMI and smoking were associated with self-reported health after adjusting for the other predictors. This gave us a stronger interpretation than relying on graphs alone.

## 15 Example Prediction for a Particular Person

This section shows how the parallel cumulative logit model can be used for one person. Our aim is to explain the practical meaning of the estimates, intercepts, exponentiation and predicted probabilities. This example follows the same manual calculation logic used in the project calculation notes (Project Calculation Notes 2026).

The example person is:

- Age = 27 years
- BMI = 10
- Gender = male
- Current smoking status = No

The BMI value of 10 is used only to reproduce the manual arithmetic example. It should not be treated as a typical adult BMI value. The purpose here is to show the calculation steps clearly.

Table 11: Illustrative person used for manual cumulative logit calculation.

| Age | BMI | Gender | Smoking |
|-----|-----|--------|---------|
| 27  | 10  | male   | No      |

### 15.1 Step 1: Extract model estimates

The parallel model has four threshold intercepts and one coefficient for each predictor. In the parallel model, the predictor coefficients are the same across the thresholds. Only the intercept changes from one threshold to another.

Table 12: Parallel model estimates used in the manual calculation.

| Term          | Estimate  |
|---------------|-----------|
| (Intercept):1 | 0.473418  |
| (Intercept):2 | 2.561695  |
| (Intercept):3 | 4.590259  |
| (Intercept):4 | 6.648358  |
| Age           | -0.011812 |
| Gendermale    | -0.153296 |
| BMI           | -0.073276 |
| SmokingYes    | -0.839791 |

Table 12 shows the estimates used in the calculation. The intercepts represent the four cumulative thresholds. The predictor estimates represent the contribution of age, gender, BMI and smoking status.

### 15.2 Step 2: Calculate the predictor part without intercept

The predictor part is calculated by multiplying each predictor value by its model estimate and then adding the results together. For this example person, male is coded as 1 and current smoking is coded as 0.

The calculation is:

$$(\beta_{Age} \times Age) + (\beta_{Male} \times Male) + (\beta_{BMI} \times BMI) + (\beta_{Smoking} \times Smoking)$$

Table 13: Manual calculation of the predictor part without intercept.

| Component            | Calculation           | Value   |
|----------------------|-----------------------|---------|
| Age part             | $-0.011812 \times 27$ | -0.3189 |
| Male part            | $-0.153296 \times 1$  | -0.1533 |
| BMI part             | $-0.073276 \times 10$ | -0.7328 |
| Smoking part         | $-0.839791 \times 0$  | 0.0000  |
| Total predictor part | Sum of all parts      | -1.2050 |

Table 13 shows the calculation of the predictor part. The total predictor part is the value before adding the threshold intercepts. This is similar to the console calculation where the age effect, gender effect, BMI effect and smoking effect are added first.

### 15.3 Step 3: Add each threshold intercept

The next step is to add each threshold intercept to the predictor part. This gives four cumulative logits, one for each threshold.

Table 14: Manual calculation after adding threshold intercepts.

| Threshold   | Intercept | Predictor_part | Logit   |
|-------------|-----------|----------------|---------|
| Threshold 1 | 0.4734    | -1.205         | -0.7316 |
| Threshold 2 | 2.5617    | -1.205         | 1.3567  |
| Threshold 3 | 4.5903    | -1.205         | 3.3853  |
| Threshold 4 | 6.6484    | -1.205         | 5.4434  |

Table 14 shows the four cumulative logits. The reason there are four logits is that the outcome has five ordered categories, so the model has four cumulative thresholds.

#### 15.4 Step 4: Apply exponentiation

The cumulative logits are on the log-odds scale. To convert each logit into cumulative odds, the exponential function is applied:

$$\text{Cumulative Odds} = \exp(\text{Cumulative Logit})$$

Table 15: Conversion from cumulative logits to cumulative odds using `exp()`.

| Threshold   | Intercept | Predictor_part | Logit   | Odds     |
|-------------|-----------|----------------|---------|----------|
| Threshold 1 | 0.4734    | -1.205         | -0.7316 | 0.4811   |
| Threshold 2 | 2.5617    | -1.205         | 1.3567  | 3.8834   |
| Threshold 3 | 4.5903    | -1.205         | 3.3853  | 29.5268  |
| Threshold 4 | 6.6484    | -1.205         | 5.4434  | 231.2270 |

Table 15 shows the result after applying `exp()`. This step explains why exponentiation is used in ordinal regression. However, these values are cumulative odds, not final category probabilities.

#### 15.5 Step 5: Convert cumulative logits to cumulative probabilities

The cumulative logit can also be converted into a cumulative probability using the logistic function:

$$P = \frac{\exp(\eta)}{1 + \exp(\eta)}$$

where  $\eta$  is the cumulative logit.

Table 16: Conversion from cumulative odds to cumulative probabilities.

| Threshold   | Intercept | Predictor_part | Logit   | Odds     | Cum_Prob |
|-------------|-----------|----------------|---------|----------|----------|
| Threshold 1 | 0.4734    | -1.205         | -0.7316 | 0.4811   | 0.3248   |
| Threshold 2 | 2.5617    | -1.205         | 1.3567  | 3.8834   | 0.7952   |
| Threshold 3 | 4.5903    | -1.205         | 3.3853  | 29.5268  | 0.9672   |
| Threshold 4 | 6.6484    | -1.205         | 5.4434  | 231.2270 | 0.9957   |

Table 16 shows the cumulative probabilities. These probabilities are cumulative, so they do not directly give the probability of one single health category. They describe the probability up to each threshold.

#### 15.6 Step 6: Calculate final category probabilities

To decide which health category is most likely for this person, the probability of each individual category must be calculated. The easiest and safest way to do this in R is to use `predict()` with `type = "response"`.

Table 17: Predicted probability of each health category for the illustrative person.

| Health    | Probability |
|-----------|-------------|
| Excellent | 0.3249      |
| Very good | 0.4704      |
| Good      | 0.1720      |
| Fair      | 0.0285      |
| Poor      | 0.0043      |

| Health | Probability |
|--------|-------------|
|--------|-------------|

Table 17 shows the predicted probability for each health category. The category with the largest predicted probability is the category the model considers most likely for this person.

Table 18: Most likely predicted health category for the illustrative person.

| Health    | Probability |
|-----------|-------------|
| Very good | 0.4704      |

Table 18 shows the most likely predicted category. This final step is important because exponentiating the logits alone does not directly decide the health category. The correct approach is to calculate the probability of each category and then identify the category with the largest predicted probability.

### 15.7 R output from the individual prediction step

After completing the manual calculation, we also checked the result using the fitted model output in R. We did this because the manual calculation shows the logic, while the model output gives the final category probabilities directly. The R output for the parallel model is shown in the next table.

Table 19: R output probabilities for the illustrative person using the parallel model.

| Health_Category | R_Output_Probability |
|-----------------|----------------------|
| Excellent       | 0.3249               |
| Very good       | 0.4704               |
| Good            | 0.1720               |
| Fair            | 0.0285               |
| Poor            | 0.0043               |

From this output, the largest probability is for the **Very good** category. This means that, for this illustrative profile, the parallel model places the person most likely in the **Very good** self-reported health category. The calculation does not mean the person must be in this category with certainty. It means this category has the highest predicted probability among the five possible health outcomes.

The same idea was then checked using the non-parallel model. The R output for the non-parallel model is shown in the next table.

Table 20: R output probabilities for the illustrative person using the non-parallel model.

| Health_Category | R_Output_Probability |
|-----------------|----------------------|
| Excellent       | 0.3409               |
| Very good       | 0.4411               |
| Good            | 0.1740               |
| Fair            | 0.0417               |
| Poor            | 0.0023               |

The non-parallel model also gives the highest probability to the **Very good** category. This was useful for us because both models gave the same final category for the example person, even though the non-parallel model used separate threshold-specific effects. This made the worked example easier to explain and also supported our final decision to keep the parallel model as the main model.

This example shows the full logic of the model in a practical way. The process is:

1. multiply each predictor value by its estimate,
2. add the threshold intercepts,
3. convert logits into odds using  $\exp()$ ,
4. convert odds into cumulative probabilities,
5. calculate individual category probabilities,
6. select the category with the highest predicted probability.

This is why estimates, odds ratios and predicted probabilities all have different roles. Estimates are used inside the model, odds ratios explain predictor effects, and predicted probabilities help explain the expected health category for a specific person.

### 15.8 Non-parallel example for the same person

The same example person can also be checked using the non-parallel model. This is useful because the non-parallel model does not use one common effect for each predictor. Instead, age, BMI, gender and smoking can have different effects at each cumulative threshold.

The example person is kept the same:

- Age = 27 years
- BMI = 10
- Gender = male
- Current smoking status = No

The non-parallel calculation is different from the parallel calculation. In the parallel model, the predictor part was the same for all thresholds. In the non-parallel model, the predictor part is different for each threshold because each threshold has its own coefficient for age, BMI, gender and smoking.

For threshold  $j$ , the calculation is:

$$\eta_j = \alpha_j + \beta_{Age,j}(\text{Age}) + \beta_{Male,j}(\text{Male}) + \beta_{BMI,j}(\text{BMI}) + \beta_{Smoking,j}(\text{SmokingYes})$$

where  $\eta_j$  is the cumulative logit for threshold  $j$ . The subscript  $j$  shows that the coefficient can change from one threshold to another.

Table 21: Non-parallel model threshold-specific estimates used in the example calculation.

| Term          | Estimate  |
|---------------|-----------|
| (Intercept):1 | 0.388767  |
| (Intercept):2 | 2.364726  |
| (Intercept):3 | 4.279083  |
| (Intercept):4 | 7.478113  |
| Age:1         | -0.006790 |
| Age:2         | -0.009686 |
| Age:3         | -0.014748 |
| Age:4         | -0.022341 |
| BMI:1         | -0.075454 |
| BMI:2         | -0.069742 |
| BMI:3         | -0.060686 |
| BMI:4         | -0.102471 |
| Gendermale:1  | -0.110064 |
| Gendermale:2  | -0.128388 |
| Gendermale:3  | -0.195104 |
| Gendermale:4  | 0.210609  |
| SmokingYes:1  | -0.809197 |
| SmokingYes:2  | -0.802619 |
| SmokingYes:3  | -0.796958 |
| SmokingYes:4  | -0.518359 |

The table above shows that the non-parallel model has more estimates than the parallel model. This is because each predictor can have a different estimate at each threshold. This made the model more flexible, but also more complex to report.

The next two tables show the non-parallel calculation. The first table shows the threshold-specific predictor parts, and the second table shows the intercepts, logits, odds and cumulative probabilities.

Table 22: Non-parallel threshold-specific predictor calculation for the illustrative person.

| Threshold   | Age_part | BMI_part | Male_part | Smoking_part | Predictor_part |
|-------------|----------|----------|-----------|--------------|----------------|
| Threshold 1 | -0.1833  | -0.7545  | -0.1101   | 0            | -1.0479        |
| Threshold 2 | -0.2615  | -0.6974  | -0.1284   | 0            | -1.0873        |
| Threshold 3 | -0.3982  | -0.6069  | -0.1951   | 0            | -1.2002        |
| Threshold 4 | -0.6032  | -1.0247  | 0.2106    | 0            | -1.4173        |

Table 23: Non-parallel logits, odds and cumulative probabilities for the illustrative person.

| Threshold   | Intercept | Logit   | Odds     | Cum_Prob |
|-------------|-----------|---------|----------|----------|
| Threshold 1 | 0.3888    | -0.6592 | 0.5173   | 0.3409   |
| Threshold 2 | 2.3647    | 1.2774  | 3.5873   | 0.7820   |
| Threshold 3 | 4.2791    | 3.0789  | 21.7349  | 0.9560   |
| Threshold 4 | 7.4781    | 6.0608  | 428.7241 | 0.9977   |

The two tables above show the manual non-parallel calculation. The important difference is that the predictor part changes across thresholds. This happens because the non-parallel model estimates separate effects for each threshold.

As with the parallel model, exponentiating the cumulative logits gives cumulative odds. The cumulative odds can then be converted to cumulative probabilities. However, these are still cumulative probabilities. To decide the most likely health category, the category probabilities must be calculated.

Table 24: Non-parallel predicted probability of each health category for the illustrative person.

| Health    | Probability |
|-----------|-------------|
| Excellent | 0.3409      |
| Very good | 0.4411      |
| Good      | 0.1740      |
| Fair      | 0.0417      |
| Poor      | 0.0023      |

The table above shows the predicted probability of each health category from the non-parallel model. The final predicted category is the category with the largest predicted probability.

Table 25: Most likely predicted health category from the non-parallel model.

| Health    | Probability |
|-----------|-------------|
| Very good | 0.4411      |

The table above gives the most likely predicted category from the non-parallel model. This shows how the non-parallel model can also be used for individual-level prediction. However, because the non-parallel model uses different effects at different thresholds, the calculation is longer and less direct to report.

This example supports the final model choice. The non-parallel model can produce predictions, but the explanation is more complicated. The parallel model is easier to explain because it uses one effect for each predictor. Since the non-parallel model did not meaningfully improve residual deviance, the simpler parallel model remains the preferred final model.

An interactive calculator was also created for the project. The file `interactive_health_category_calculator.html` allows the user to enter age, BMI, gender and smoking status and then check the predicted probabilities and the most likely health category for both models.

## Graphical Representation of Person-Level Prediction

The individual prediction example can also be summarised using a graph and tables. This section uses the same illustrative person from the manual calculation: age 27 years, BMI 10, gender male and current smoking status No. A non-interactive version of the graph is included in the thesis PDF so that the result is visible in the report. The interactive version is implemented using Shiny and must be run from the RMarkdown code file in RStudio. It will not appear as a fully working interactive graph in an already rendered static HTML file. Running the code allows users to change age, BMI, gender and smoking status and view the updated predicted probabilities.

Table 26: Input values used for the person-level prediction summary.

| Input          | Value |
|----------------|-------|
| Age            | 27    |
| BMI            | 10    |
| Gender         | male  |
| Smoking status | No    |

Table 27: Predicted probabilities for the illustrative person under the parallel and non-parallel models.

| Model        | Health_Category | Predicted_Probability |
|--------------|-----------------|-----------------------|
| Parallel     | Excellent       | 0.3249                |
| Parallel     | Very good       | 0.4704                |
| Parallel     | Good            | 0.1720                |
| Parallel     | Fair            | 0.0285                |
| Parallel     | Poor            | 0.0043                |
| Non-parallel | Excellent       | 0.3409                |
| Non-parallel | Very good       | 0.4411                |
| Non-parallel | Good            | 0.1740                |
| Non-parallel | Fair            | 0.0417                |
| Non-parallel | Poor            | 0.0023                |

Table 28: Most likely predicted health category for the illustrative person.

| Model        | Predicted_Category | Predicted_Probability |
|--------------|--------------------|-----------------------|
| Non-parallel | Very good          | 0.4411                |
| Parallel     | Very good          | 0.4704                |



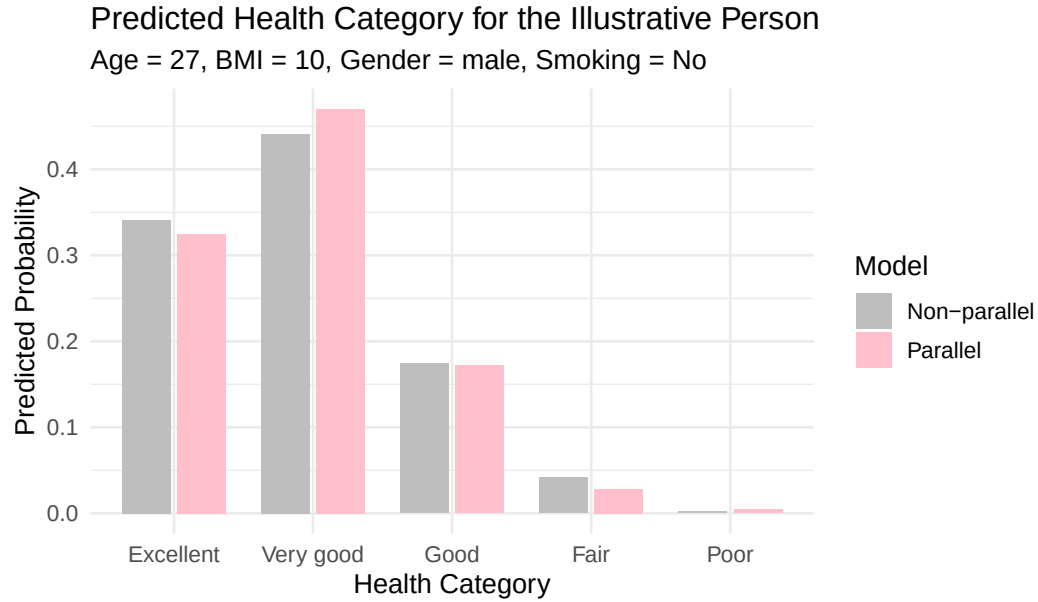


Figure 8: Predicted health-category probabilities for the illustrative person under the parallel and non-parallel models.

The table and graph show that both models predict the same most likely category, Very good. The parallel model gives a predicted probability of 0.4704 for Very good, while the non-parallel model gives 0.4411. This means that even though the non-parallel model is more flexible, it does not change the final predicted category for this illustrative person.

## 16 Predicted Probabilities

Predicted probabilities are useful because they show the model results in an easier form than log-odds or odds ratios. Instead of only saying that age or smoking changes the odds, predicted probabilities show how the probability of each health category changes across a predictor.

In our project, predicted probabilities were calculated across the observed age range. BMI was fixed at the mean value, gender was fixed at the reference category, and smoking was fixed as No. This makes the age pattern easier to view.

The following figure shows the predicted probabilities from the parallel model.

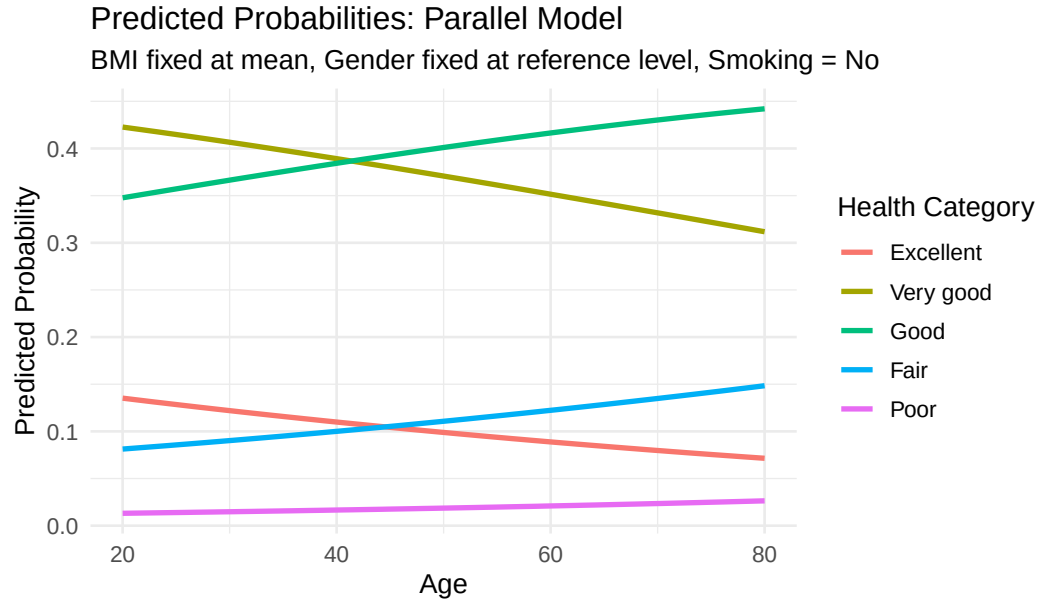


Figure 9: Predicted probabilities from the parallel model.

The figure shows how the predicted probabilities change with age under the parallel model. The graph is easier to understand than log-odds because it shows probabilities for each health category directly.

The following figure shows the predicted probabilities from the non-parallel model.

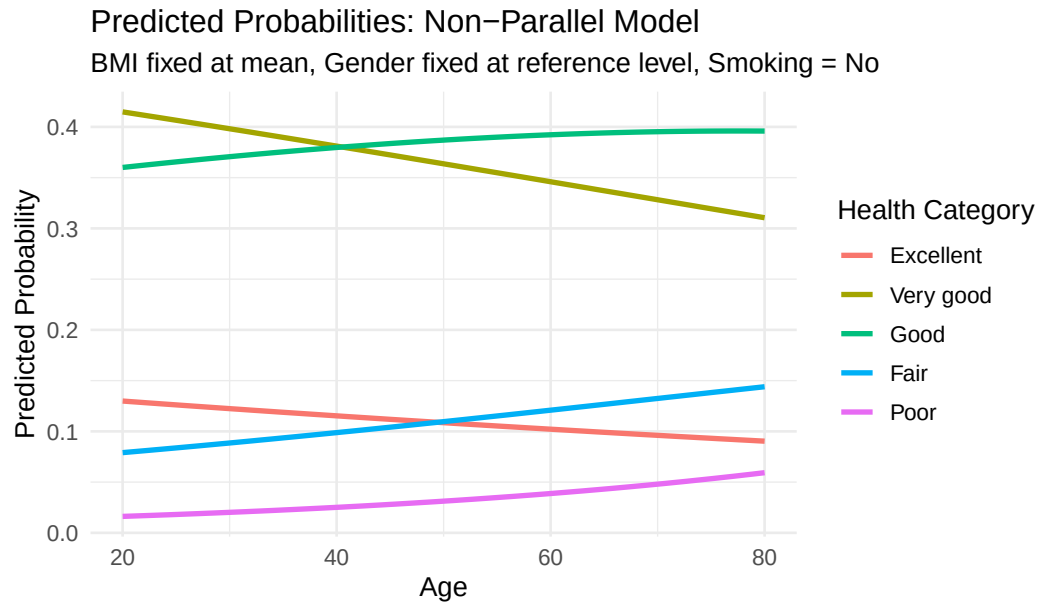


Figure 10: Predicted probabilities from the non-parallel model.

The figure shows the predicted probability pattern under the non-parallel model. This model is more flexible, so the curves may differ slightly from the parallel model. However, flexibility alone is not enough to choose it as the final model.

The following figure compares the predicted probabilities from the parallel and non-parallel models side by side.

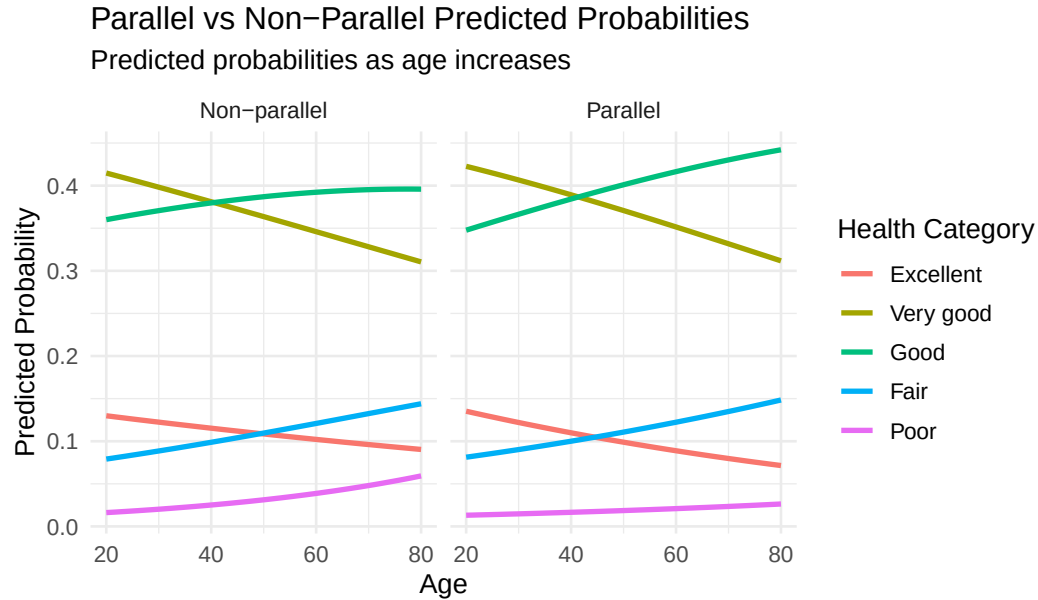


Figure 11: Parallel and non-parallel predicted probabilities compared.

The figure shows that the predicted probability patterns from the two models are broadly similar. This supported our decision to keep the parallel model as the main model. The non-parallel model did not produce a large enough improvement to justify the extra complexity.

## 17 How to Read the Predicted Probability Curves

The predicted probability curves show how the model behaves across the observed age range. These curves are useful because they translate the model output into probabilities. This makes the results easier to understand than log-odds or odds ratios alone.

Each line in the predicted probability graph represents one health category. The x-axis shows age and the y-axis shows the predicted probability. BMI is fixed at the mean value, gender is fixed at the reference category and smoking is fixed as No. This setup allows the graph to focus on the relationship between age and self-reported health while keeping the other predictors constant.

If the predicted probability for a category increases as age increases, it means the model expects that category to become more likely with age under the fixed profile. If the probability decreases, it means that category becomes less likely with age. In this project, the predicted probability patterns support the idea that age is associated with changes in self-reported health categories.

The predicted probability curves are not meant to describe every possible person. They describe one controlled profile. Different curves would be obtained if BMI, gender or smoking status were changed. For example, the predicted probabilities for a current smoker may be different from the predicted probabilities for a non-smoker. This is why predicted probabilities are useful for individual profiles as well as for general interpretation.

Our comparison between the parallel and non-parallel predicted probability plots is also important. The non-parallel model is more flexible, but the overall probability patterns were broadly similar to the parallel model. This supports the final decision to use the simpler parallel model as the main model.

## 18 Model Comparison and Final Model Choice

We based the final model choice on both statistical results and interpretation. The non-parallel model had a slightly lower residual deviance than the parallel model. However, the reduction was only 1.068. Since the residual deviance values were above 7400, this difference was very small. This follows a common modelling

principle: a more complex model should only be preferred when it provides a meaningful improvement in fit or interpretation (Course Material 2026b; Burnham and Anderson 2002).

The parallel model has several advantages. First, it gives one coefficient and one odds ratio for each predictor. This makes it clearer to report the effect of age, BMI, gender and smoking. Second, it produced stable fit statistics such as residual deviance, log-likelihood, AIC and BIC. Third, the predicted probability patterns were similar to the non-parallel model.

The non-parallel model was useful as a sensitivity check. It showed that allowing threshold-specific effects did not meaningfully improve the model. However, the output was more complex because each predictor had separate coefficients at different thresholds. This made the model more complex to report clearly.

We summarise the final model decision in Table 23.

Table 29: Final model decision summary.

| Item            | Explanation                                                                 |
|-----------------|-----------------------------------------------------------------------------|
| Preferred model | Parallel cumulative logit model                                             |
| Reason 1        | The non-parallel model reduced residual deviance by only about 1.068.       |
| Reason 2        | The predicted probability patterns were broadly similar across both models. |
| Reason 3        | The parallel model was simpler, more stable and clearer to report.          |

Based on the model comparison, the parallel cumulative logit model was selected as the main model for this project. It gave us a useful balance between statistical fit, stability and interpretability.

## 19 Interactive Visualisation

Two interactive outputs were prepared for this project.

The first output is an interactive predicted probability graph saved as:

`interactive_predicted_probabilities.html`

This file allows the predicted probability curves to be explored by hovering over the lines.

The second output is an interactive calculator saved as:

`interactive_health_category_calculator.html`

This calculator allows a user to enter age, BMI, gender and smoking status and then view the predicted probabilities for each health category. It also identifies the most likely predicted health category. The calculator displays results for both the parallel and non-parallel models so that the fitted outputs can be compared for a particular person.

These files are included with the project materials and should be uploaded to the GitHub repository together with the report and code files:

[https://github.com/manasvisait1504/Ordinal\\_regression\\_Models](https://github.com/manasvisait1504/Ordinal_regression_Models)

The PDF report cannot display the interactive HTML objects directly, so we include their file names and explains their purpose. The static figures and tables in the report provide the printed version of the results, while the interactive files provide an additional way to explore the model outputs.

## 20 Source Use and Referencing

The report uses a separate BibTeX bibliography file named `references.bib`. This file is called in the YAML header of the RMarkdown document using `bibliography: references.bib`. The citations in the text identify the sources used for theory, methods, data, software and interpretation.

The main statistical theory for ordinal regression is supported by standard ordinal regression references, including McCullagh and Agresti (McCullagh 1980; Agresti 2010). The discussion of ordered regression

models, parallel models and non-parallel alternatives is also supported by Fullerton and Xu (Fullerton and Xu 2016) and the uploaded ordered regression course material (Course Material 2026a, 2026c).

The dataset source is identified through NHANES documentation and the R package used to access the data (Centers for Disease Control and Prevention, National Center for Health Statistics 2024a, 2024b; Pruim 2016). The software sources are also cited, including R, RMarkdown, VGAM, ggplot2, knitr, plotly and htmlwidgets (R Core Team 2024; Allaire, Xie, Dervieux, et al. 2024; Yee 2024; Wickham 2016; Xie 2015; Sievert et al. 2024; Vaidyanathan et al. 2024). Model-fit concepts such as log-likelihood, AIC and residual deviance are supported by the likelihood and deviance notes and model-selection references (Course Material 2026b; Burnham and Anderson 2002).

Grammarly was used only for grammar, spelling and writing clarity checks during the final editing of the report. It was not used for statistical analysis, model fitting, interpretation of results, or generation of the research findings.

## 21 Code Availability

The RMarkdown code, analysis files and project materials are available in the GitHub repository:

[https://github.com/manasvisait1504/Ordinal\\_regression\\_Models](https://github.com/manasvisait1504/Ordinal_regression_Models)

The repository includes the documented analysis code used for data preparation, exploratory analysis, model fitting, model comparison, individual prediction examples and predicted probability plots. The PDF report does not include the analysis code; it only includes the model outputs, tables, figures and explanation required for the report.

Two interactive HTML files should also be included in the repository:

`interactive_predicted_probabilities.html`

and

`interactive_health_category_calculator.html`

The first file provides an interactive predicted probability graph. The second file provides an interactive person-level calculator where the user can enter age, BMI, gender and smoking status and check which health category is most likely under the fitted models.

## 22 Discussion

Our results show that ordinal regression is a suitable method for analysing self-reported health because the outcome is ordered. The analysis also shows that the selected predictors have meaningful relationships with self-reported health.

Age and BMI both had negative coefficients in the parallel model. This suggests that older age and higher BMI are associated with lower odds of better self-reported health. This is consistent with the exploratory graphs, where poorer health categories tended to include older participants and higher BMI values.

Current smoking had the strongest association among the predictors. The odds ratio for current smokers was below 1, suggesting that current smokers had lower odds of reporting better health compared with non-smokers, after adjusting for age, BMI and gender. This result is important because smoking is a major lifestyle factor related to health.

Gender was also included as a predictor. The coefficient for male participants compared with the reference category was negative in the parallel model. This suggests a difference in self-reported health by gender after adjustment for the other variables.

Our comparison between the parallel and non-parallel models was an important part of the analysis. The non-parallel model was more flexible, but it did not substantially improve model fit. This supported our use of the simpler parallel model. In applied data analysis, the best model is not always the most complex model. A model should also be understandable, stable and useful for answering the research question.

A further strength of the report is that it includes both model-level interpretation and person-level prediction. The model-level interpretation explains the general association of age, BMI, gender and smoking with self-reported health. The person-level prediction example shows how the fitted model can be used for a specific profile. This makes the analysis easier to understand because it connects the statistical output to a practical calculation.

When we first compared the two models, we were surprised that the more flexible non-parallel model did not give a meaningful improvement. We expected that allowing separate effects at each threshold might improve the model more clearly. Instead, the predicted probability patterns were broadly similar and the residual deviance improvement was very small. This helped us understand that model choice is not only about flexibility, but also about whether the model gives a stable and understandable answer.

## 23 Limitations

This project has several limitations.

First, the analysis uses observational survey data. This means the results should be interpreted as associations, not causal effects. For example, the model can show that smoking is associated with poorer self-reported health, but it cannot prove causation from this analysis alone.

Second, the outcome variable is self-reported. Self-reported health is useful, but it is subjective. Different people may interpret the health categories differently. One person may describe their health as Good while another person with similar health conditions may choose Fair.

Third, only a small number of predictors were included. Age, BMI, gender and smoking are important, but many other variables may also affect self-reported health. These include income, education, physical activity, chronic disease, mental health, access to healthcare and diet.

Fourth, the project used complete cases after selecting the variables. Although the final modelling dataset had no missing values for the selected variables, complete-case analysis may still exclude some records from the original NHANES data.

Fifth, the analysis did not apply the full complex survey design features of NHANES, such as sampling weights, strata and clusters. Including survey design features would make the analysis more representative of the United States population and would follow the full survey-analysis framework recommended for NHANES data (Centers for Disease Control and Prevention, National Center for Health Statistics 2024a, 2024b).

Finally, the non-parallel model was more complex and harder to interpret. Although it was useful as a sensitivity check, its additional complexity was not justified by the very small reduction in residual deviance.

## 24 Conclusion and Future Work

This project analysed self-reported general health using cumulative logit ordinal regression. The outcome variable was ordered from Excellent to Poor, so ordinal regression was more appropriate than ordinary linear regression. The predictors included age, BMI, gender and current smoking status.

The exploratory analysis showed clear patterns in the data. Most participants reported Good or Very good health, while fewer participants reported Poor health. Age and BMI also showed visible differences across health categories.

The parallel cumulative logit model showed that age, BMI, gender and smoking were associated with self-reported health. Odds ratios below 1 indicated lower cumulative odds of better self-reported health. Current smoking showed the strongest association with poorer self-reported health.

We fitted the non-parallel model as a sensitivity check. It produced a slightly lower residual deviance than the parallel model, but the difference was only 1.068. This was too small to justify the extra complexity. Therefore, the parallel model was selected as the final model because it was simpler, more stable and clearer to report.

Future work could extend this project in several ways. First, additional predictors such as income, education, physical activity, chronic illness and mental health could be included. Second, NHANES survey weights and design variables could be used to make the results more representative. Third, interaction effects could be tested, for example whether the association between BMI and self-reported health differs by gender or

smoking status. Fourth, partial proportional odds models could be explored as a middle option between the parallel and non-parallel models. Finally, an interactive dashboard could be developed to allow users to explore predicted probabilities for different age, BMI, gender and smoking profiles.

Our project code and interactive predicted probability visualisation are available through the project repository: [https://github.com/manasvisait1504/Ordinal\\_regression\\_Models](https://github.com/manasvisait1504/Ordinal_regression_Models). Overall, this project shows that cumulative logit ordinal regression is a useful and interpretable method for analysing ordered health outcomes. It also shows that model choice should consider both statistical fit and practical interpretation.

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